

Combining photovoltaic and electrochemical conversions within a photocathode for CO₂ reduction.

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Context

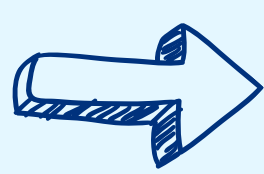
420 ppm

Current atmospheric CO₂ concentration

The reaction has a ΔG° of 1.33 eV/e⁻, motivating the choice of CIGS ($E_g \approx 1.2$ eV) as the main PV absorber within the multilayer stack.

Achieving the optimal design of multi-layer photocathodes requires an iterative process between experimentation and modeling:

- Experiments are conducted by laboratories involved in the ANR project GASPER in order to compare the model with experimental results.
- Photocathodes involve two coupled parts: a photovoltaic part and an electrochemical part, each raising specific modeling challenges.



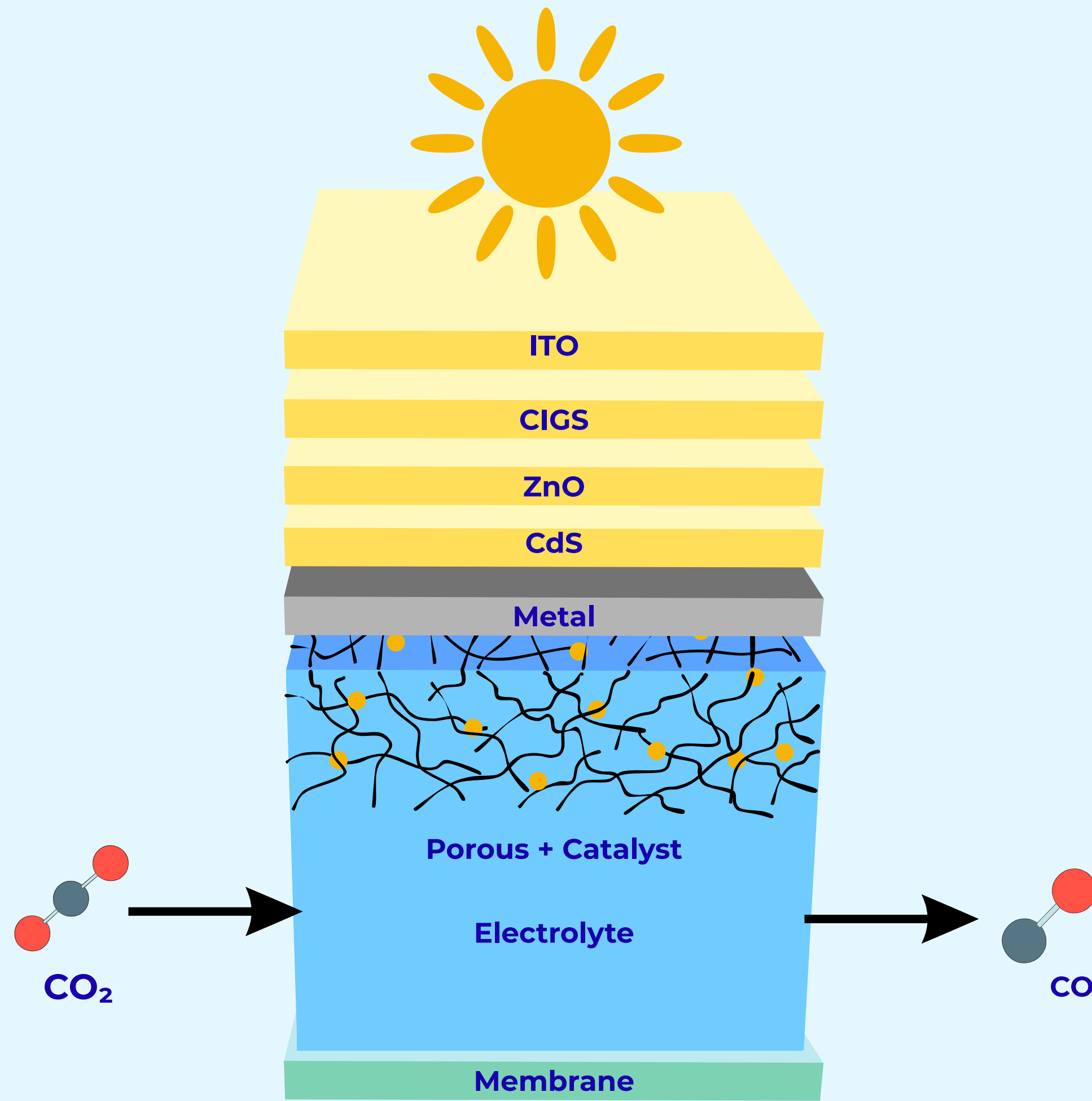
A model of each part is first developed and compared with experimental results, before coupling them into a full device model and ultimately guiding the optimization of layer composition and thickness (inverse design).

Anode



Syngas

Cathode



Model

To model the full device, several approaches are considered for each sub-part as outlined below.

Photocathode model

Photovoltaic Part

Charge transport

- Drift-Diffusion-Reaction^[1] (Local balance)
- Photogeneration: Electromagnetism and/or Radiative Transfer

Detailed balance^[3] (System approach)

Electrochemical part

Kinetic

- Butler-Volmer/Tafel^[2]
- Marcus...
- Ad hoc model/DFT

CO₂ mass transfer

- Gas diffusion electrode
- Dissolved gas in the electrolyte

Geometry^[2]

1D homogeneous

3D heterogeneous

Optimized 3D design

The whole model will be solved with a custom deterministic in-house code.

Preliminary work

6 months of thesis

A first model of the PV part with the charge transport approach has been implemented, including p-n junction simulation.

- The p-n junction simulation, without carrier generation, successfully reproduces results from the literature^[1], providing electric field and carrier profiles across the device.
- A first approach to photogeneration has been implemented using a Bouguer absorption profile.

For the electrochemical part:

- Standard Butler-Volmer kinetics are commonly used but their validity in this configuration remains questionable.
- Several diffusion models have been reviewed, and the role of the electric double layer is currently under investigation.

The in-house code is motivated by an appropriation objective, enabling future coupling with the electrochemical part, unlike existing PV simulators.

Perspectives

Short-term

- Implementation of carrier generation under solar illumination
- Extension of the PV model to a multi-layer structure (CIGS/ZnO/CdS)
- Development of a complete electrochemical model for the porous cathode, including interfacial reactions and species transport

Mid-term

- Coupling of the PV and catalysis parts into a full device model
- Validation of the model against experimental data
- Comparison of the different modeling approaches

Long-term

- Optimization of the photocathode layer composition, thickness and nanostructuration to maximize CO₂ reduction efficiency
- Complete inverse design of the photocathode

References

- [1] P. Farrell, N. Rotundo, D.H. Doan, M. Kantner, J. Fuhrmann, Th. Koprucki, « Drift-Diffusion Models », in Handbook of Optoelectronic Device Modeling and Simulation, vol. 2, J. Piprek (éd.), CRC Press, 2017, pp. 733–771.
- [2] Le, T. D., D. Lasseux, L. Zhang, et al. « Multiscale Modelling of Diffusion and Enzymatic Reaction in Porous Electrodes in Direct Electron Transfer Mode ». Chemical Engineering Science 248 (février 2022): 117157. <https://doi.org/10.1016/j.ces.2021.117157>.
- [3] Vossier, Alexis, Federico Gualdi, Alain Dollet, Richard Ares, et Vincent Aimez. « Approaching the Shockley-Queisser Limit: General Assessment of the Main Limiting Mechanisms in Photovoltaic Cells ». Journal of Applied Physics 117, n° 1 (2015): 015102. <https://doi.org/10.1063/1.4905277>.

